

A memory construction

 committed events
 open-day buffer



day closes

LLM trajectory extraction
typed facts and citations

walking, 18:22–18:48 [e_{17}, e_{18}]
✓ source-supported facts → W_t
✗ unsupported distance rejected

LLM regularity proposal
7 completed records

trigger weekday, low activity after 17:30
expect walk within 18:00–19:00
replay **M T W T F S S** excluded
replay gate passed → active

ProactMem

event
memory
 S_t

72 h, ≤ 99 events

temporal
trajectories
 W_t

source-checked

behavioral
regularities
 L_t

replay-assessed

B event-driven decision

 **memory preparation**

retrieve prior memory; match current conditions
top 3 regularities and observable service state

semantic screening

memory relevance
event-local evidence

escalate

context C_t



e_t 18:30, low activity



decision LLM

$\pi_\theta(C_t)$

service assessment



intervention control

filter repeated proposals

intervene

“Short walk?”

abstain

commit e_t after finalization